MACHINE LEARNING-BASED MULTICLASS NETWORK INTRUSION DETECTION

An Intelligent Cybersecurity Classification Approach
Using Network Traffic Features

FINAL PROJECT REPORT

DATA RECORDS	FEATURES	TRAFFIC CLASSES
210,736	**70**	**12**

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Project Snapshot

This report documents a supervised machine-learning experiment on multiclass network traffic classification. It summarizes the prepared dataset, model comparison, detailed evaluation, and the most important observed limitations.

Highest accuracy	Highest Macro F1	XSS F1-score
99.57%	**92.69%**	**37.17%**
XGBoost	**Random Forest**	**XGBoost**

Abstract

This project evaluates supervised machine learning for multiclass network intrusion detection using network-traffic features. The modeling dataset contains 210,736 records, 70 predictive features, and 12 traffic classes. Logistic Regression, Random Forest, and XGBoost were compared using accuracy, macro precision, macro recall, and macro F1-score. XGBoost achieved the highest overall accuracy at 99.57%, while Random Forest achieved the highest Macro F1-score at 92.69%. Class-level analysis revealed a notable weakness in Web Attack - XSS detection: the class achieved an F1-score of 37.17%, with 42 of 130 test examples correctly classified. Many other classes were classified with very high performance. These results illustrate why overall accuracy should be complemented by macro-averaged metrics and per-class error analysis when evaluating security-related classifiers.

1. Introduction

Network environments generate large volumes of traffic, including benign activity and flows associated with malicious behavior. Intrusion Detection Systems (IDS) support security monitoring by identifying traffic patterns that may indicate attacks. Supervised machine learning can learn associations between network-flow features and labeled traffic categories, offering a data-driven approach to traffic classification.

This project treats intrusion detection as a multiclass classification problem rather than a binary benign/malicious decision. This enables comparison across attack categories and highlights where model performance is strong or requires further work.

2. Problem Statement

The objective is to evaluate models that classify network traffic into 12 labels. The central challenge is to measure both overall predictive performance and the ability to detect each class, including classes with relatively few examples. High aggregate accuracy can conceal weak performance on a less represented but security-relevant attack type.

3. Objectives

- Prepare network-traffic data for supervised multiclass classification.
- Explore the distribution of the 12 traffic classes.
- Train and compare Logistic Regression, Random Forest, and XGBoost.
- Evaluate models using Accuracy, Macro Precision, Macro Recall, and Macro F1-score.
- Analyze XGBoost class-level performance and confusion patterns.

• Identify influential predictors using XGBoost feature importance.

• Document limitations and propose practical future improvements.

4. Dataset Description

The prepared modeling dataset contains 210,736 records and 70 predictive features, with the target column named Label. The 12 classes are BENIGN, Bot, DDoS, DoS GoldenEye, DoS Hulk, DoS Slowhttptest, DoS slowloris, FTP-Patator, PortScan, SSH-Patator, Web Attack - Brute Force, and Web Attack - XSS.

Class representation varies substantially. DDoS and BENIGN are the largest groups, while Web Attack - Brute Force and Web Attack - XSS are comparatively small. This distribution matters when interpreting macro-averaged metrics and minority-class behavior.

Traffic class	Records	Share
DDoS	49,999	23.73%
BENIGN	49,232	23.36%
PortScan	39,388	18.69%
DoS Hulk	37,991	18.03%
DoS GoldenEye	10,286	4.88%
FTP-Patator	5,933	2.82%
DoS slowloris	5,385	2.56%
DoS Slowhttptest	5,228	2.48%
SSH-Patator	3,219	1.53%
Bot	1,953	0.93%
Web Attack - Brute Force	1,470	0.70%
Web Attack - XSS	652	0.31%

5. Data Preprocessing

The workflow included dataset inspection, quality assessment, train-test separation, missing-value handling, and feature scaling where appropriate. Target labels were encoded for model training and evaluation. The prepared feature matrix contained 70 predictors. The reported split consisted of 168,588 training records and 42,148 test records.

Preprocessing operations should be fitted on training data and then applied to test data to avoid leakage. The reported metrics represent performance on the held-out test split used in the notebook.

6. Model Development

6.1 Logistic Regression

Logistic Regression served as a linear baseline and provides a useful comparison against nonlinear ensemble methods.

6.2 Random Forest

Random Forest combines multiple decision trees and can model nonlinear relationships and interactions among features. It delivered the strongest Macro F1-score among the evaluated models.

6.3 XGBoost

XGBoost is a gradient-boosted decision-tree method that builds an ensemble sequentially to improve predictions. It achieved the highest overall accuracy in this experiment, but class-level results show that performance was not uniform.

7. Model Performance Comparison

Model	Accuracy	Macro Precision	Macro Recall	Macro F1
Logistic Regression	97.14%	94.22%	83.36%	83.47%
Random Forest	99.52%	92.94%	92.64%	92.69%
XGBoost	99.57%	92.86%	92.65%	92.66%

Interpretation: XGBoost had the highest accuracy (99.57%), while Random Forest had the highest Macro F1-score (92.69%), narrowly exceeding XGBoost (92.66%). Macro F1 gives each class equal weight, making it useful when class representation varies.

8. Detailed XGBoost Evaluation

XGBoost classified many classes with near-perfect performance. Strong results included DDoS, major DoS categories, PortScan, FTP-Patator, and SSH-Patator. Bot also achieved a high F1-score of 98.98%. The main weakness was in the web-attack classes.

Class	Precision	Recall	F1-score	Support
BENIGN	99.88%	99.87%	99.87%	9,846
Bot	98.73%	99.23%	98.98%	391
DDoS	99.99%	99.99%	99.99%	10,000
DoS GoldenEye	99.95%	99.90%	99.93%	2,057
DoS Hulk	99.93%	99.92%	99.93%	7,598
DoS Slowhttptest	99.71%	99.33%	99.52%	1,046
DoS slowloris	99.54%	99.63%	99.58%	1,077
FTP-Patator	100.00%	100.00%	100.00%	1,187
PortScan	99.94%	99.97%	99.96%	7,878
SSH-Patator	100.00%	100.00%	100.00%	644
Web Attack - Brute Force	72.95%	81.63%	77.05%	294
Web Attack - XSS	43.75%	32.31%	37.17%	130

Critical limitation: Web Attack - XSS had an F1-score of 37.17%. Only 42 of 130 test examples were correctly classified, so this category requires targeted improvement before any practical security use.

9. Confusion Matrix Analysis

The confusion matrix showed very few errors for many classes. The clearest weakness was confusion between Web Attack - XSS and Web Attack - Brute Force. Of the 130 XSS test examples, 42 were correctly classified and 85 were predicted as Brute Force. In the other direction, 50 of the 294 Brute Force examples were predicted as XSS.

This two-way confusion may indicate that the available flow-level features do not sufficiently distinguish these web-attack categories. It motivates additional web-application-specific features and targeted error analysis.

10. Feature Importance

XGBoost feature importance ranked Destination Port as the most influential feature, followed by TCP/window-related fields, packet-rate measures, packet-length statistics, inter-arrival-time measures, and flow-duration features.

Rank	Feature	Importance
1	Destination Port	0.070835
2	Init_Win_bytes_backward	0.050149
3	Bwd Packets/s	0.033387
4	Fwd Packet Length Max	0.029279
5	Bwd Header Length	0.026665
6	Flow IAT Mean	0.025531
7	Flow IAT Std	0.024603
8	Flow Duration	0.024454
9	Init_Win_bytes_forward	0.023757
10	Flow Packets/s	0.023549

Other influential features included Flow IAT Max, Packet Length Mean, min_seg_size_forward, Fwd IAT Mean, Average Packet Size, Fwd IAT Max, Fwd Header Length.1, Fwd Packets/s, Avg Bwd Segment Size, and Bwd Packet Length Mean. Feature importance indicates how the trained model uses predictors; it does not establish causality.

11. Key Findings

- Tree-based ensemble methods substantially outperformed Logistic Regression on the overall task.
- XGBoost achieved the highest accuracy at 99.57%.
- Random Forest achieved the highest Macro F1-score at 92.69%.
- Most classes were detected with very high class-level performance.
- Web Attack - XSS was the clearest weakness, with F1-score 37.17%.
- XSS and Brute Force exhibited strong two-way confusion.

• Destination Port, packet-rate, packet-size, timing, TCP-window, and flow-duration features were influential.

• Macro and class-level metrics revealed weaknesses that accuracy alone would obscure.

12. Limitations

• The evaluation uses one prepared dataset and one held-out test split; real-world generalization is not established.

• Class representation varies substantially, and the XSS test support is relatively small (130 examples).

• Feature importance is model-dependent and should not be interpreted causally.

• The project evaluates offline classification rather than a deployed real-time IDS.

• Further hyperparameter optimization, robustness testing, and independent validation may strengthen confidence in the results.

13. Future Work

• Evaluate class weighting, resampling, and other approaches for underrepresented classes.

• Explore web-application-specific features to better separate XSS and Brute Force traffic.

• Validate on independent datasets and newer network environments.

• Perform systematic hyperparameter tuning and robustness tests.

• Use explainability methods such as SHAP to investigate feature contributions and individual predictions.

• Prototype a real-time inference pipeline and monitoring dashboard.

14. Conclusion

This project demonstrates that machine learning can classify the prepared network-traffic dataset with very high aggregate performance. XGBoost achieved 99.57% accuracy, while Random Forest achieved the highest Macro F1-score at 92.69%. However, class-level analysis shows non-uniform performance: Web Attack - XSS achieved only 37.17% F1-score and was frequently confused with Web Attack - Brute Force.

The key conclusion is that a credible intrusion-detection model must be evaluated beyond accuracy. Macro-averaged metrics, per-class scores, and confusion analysis are essential for identifying weaknesses that matter in security settings. Future work should focus on minority-class detection, additional features, independent validation, explainability, and deployment-oriented evaluation.

15. References and Reproducibility

The exact dataset publication or repository should be cited from the source used in the notebook. This report does not invent dataset authors, publication details, or a DOI that were not confirmed in the project notes.

Recommended software documentation:

• Scikit-learn User Guide: https://scikit-learn.org/stable/user_guide.html

• XGBoost Documentation: https://xgboost.readthedocs.io/

For a final academic submission, add the precise dataset citation and access date from the original dataset page.

END OF REPORT